

Test: Design of PID Controllers

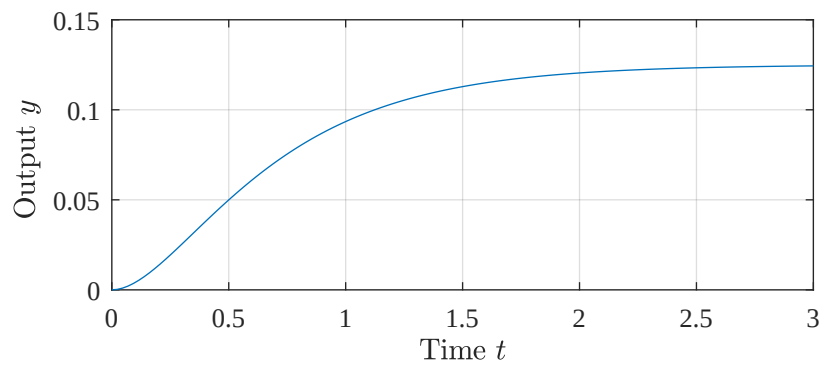
1. Include a PID-controller in the following block diagram.



2. Derive the closed-loop transfer function of a system $G(s) = \frac{1}{\tau s + 1}$ with a P-control with gain K_p .
3. Include a P-controller with gain K_p in the following block diagram in addition to a feedforward, and compute an expression for the feedforward such that the steady state error is eliminated.



4. Consider the following step response of an open-loop system. Use Ziegler-Nichols tuning method to provide initial guesses for a PI-controller for the system.



$$K_p =$$

$$K_i =$$